

College of Engineering

Electrical & Computer Engineering

18-738 Sports Technology

Auto-Negotiate: Autonomous Negotiation Agent

Team 10

Abdullah Almansour

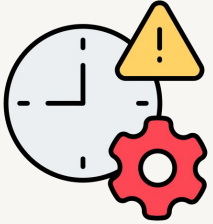
Madhavi Gulavani

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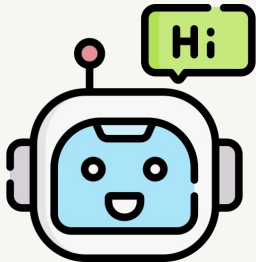
Problem Statement



Negotiation in nature is difficult, let alone doing it in a rapidly changing sector
Human as negotiator must be well prepared and informed for the best possible outcome

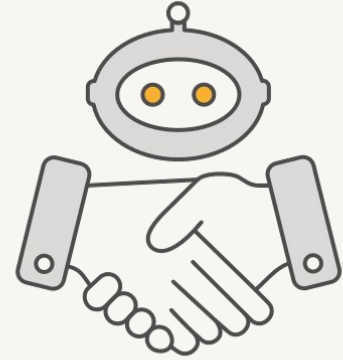


Current tools in the [Sports sector](#) are negotiator assistant, e.g., providing data and contracts revisions.
There is a potential for developing a system that automate the process entirely, especially in [Sports sector](#)



[Chatbot as a negotiator agent](#) for negotiating salaries, contract benefits, and transfers for players, coaches, etc.
The current AI advancement will significantly facilitate the progress in this direction

The Why: Auto-Negotiation in Sports



Innovative Autonomy

Uses Agentic AI and reasoning LLMs to conduct full negotiations, unlike passive chatbots. It secures best deals by removing human bias and fatigue.

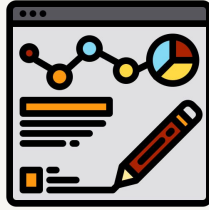
Competitive Edge

Vertical specialization outperforms generic tools like [Pactum \(procurement\)](#) or [Zone7 \(analytics\)](#). Eliminates the need for complex prompt engineering.

Market Viability

Fills a critical gap for 24/7 autonomous negotiation in a high-stakes, dynamic sector where human agents face bandwidth and data-processing limits.

Build + Data



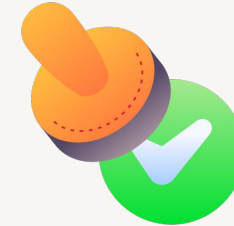
Model

- **Multi-agent:** Club/GM ↔ Player/Agent (+ optional League/Rules Referee)
- Offers as a **structured term sheet** (years, salary, bonuses, clauses)
- **Constraint checker** for cap/rules + auto-fix suggestions
- **RAG grounding:** market comps + rules/clauses snippets
- **Negotiation policy:** goals, concessions, walk-away

Data

- **Historical deals:** contracts, renewals, transfers + outcomes
- **Rules corpus:** CBA/league rules + clause templates
- **Self-play transcripts:** synthetic negotiations under constraints

Validation + Demo



Validation

- **Legality/compliance rate** (cap + rules)
- **Market realism** (distance to comparable deals)
- **Efficiency** (rounds-to-agreement)
- **Outcome quality** (value for both sides vs baseline)

Baselines

- **Single-prompt LLM negotiator** (no tools)
- **LLM + retrieval only** (no constraint checker)

Demo

- Enter scenario (player + team constraints + priorities)
- Watch live **offer timeline + term sheets**
- Export final contract + short “why” notes + citation

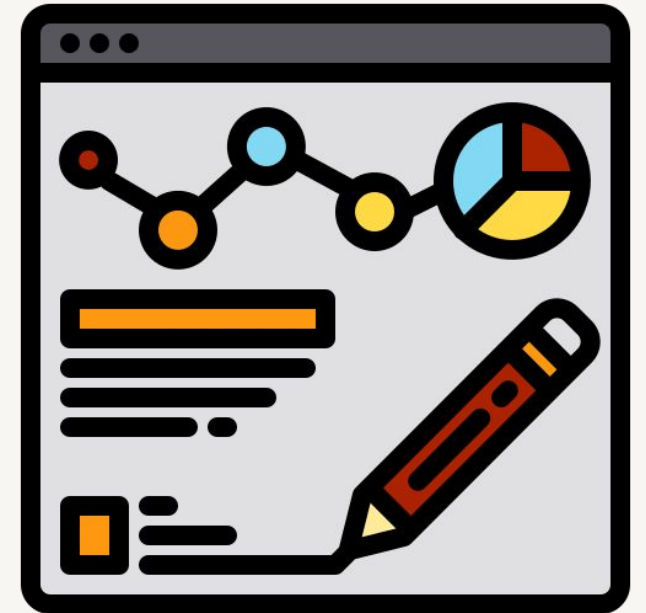
The How: *Build + Data*

Model

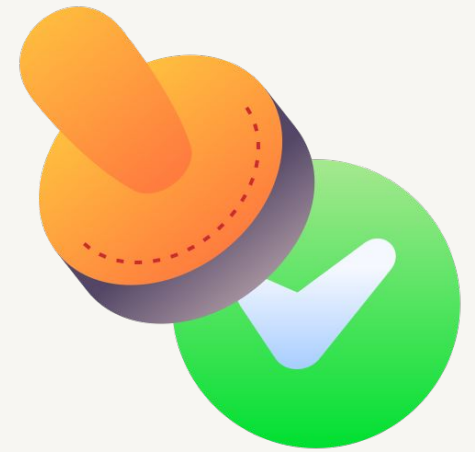
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The How: *Validate + Demo*



Validation

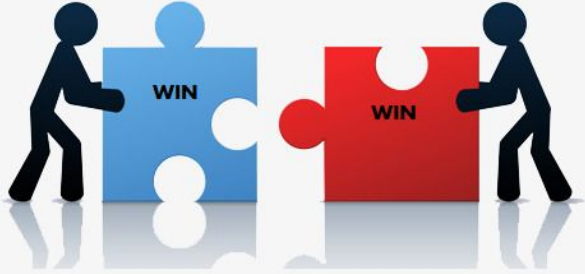
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Auto-Negotiate: Autonomous Negotiation Agent

Team 10: Current progress

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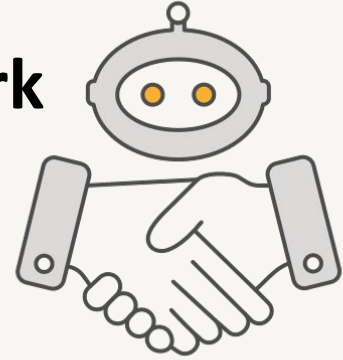
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Modular Multi-agent Negotiation Framework



Multi-Agent Negotiation Engine Backbone

LangGraph-orchestrated agents (Club, Player, League/Rules) coordinate through A2A messaging to simulate real negotiation dynamics and generate structured offers.

Salary Prediction & Market Intelligence

ML-based salary predictor estimates fair market value using player performance, historical deals, and comparable contracts to anchor negotiation strategies.

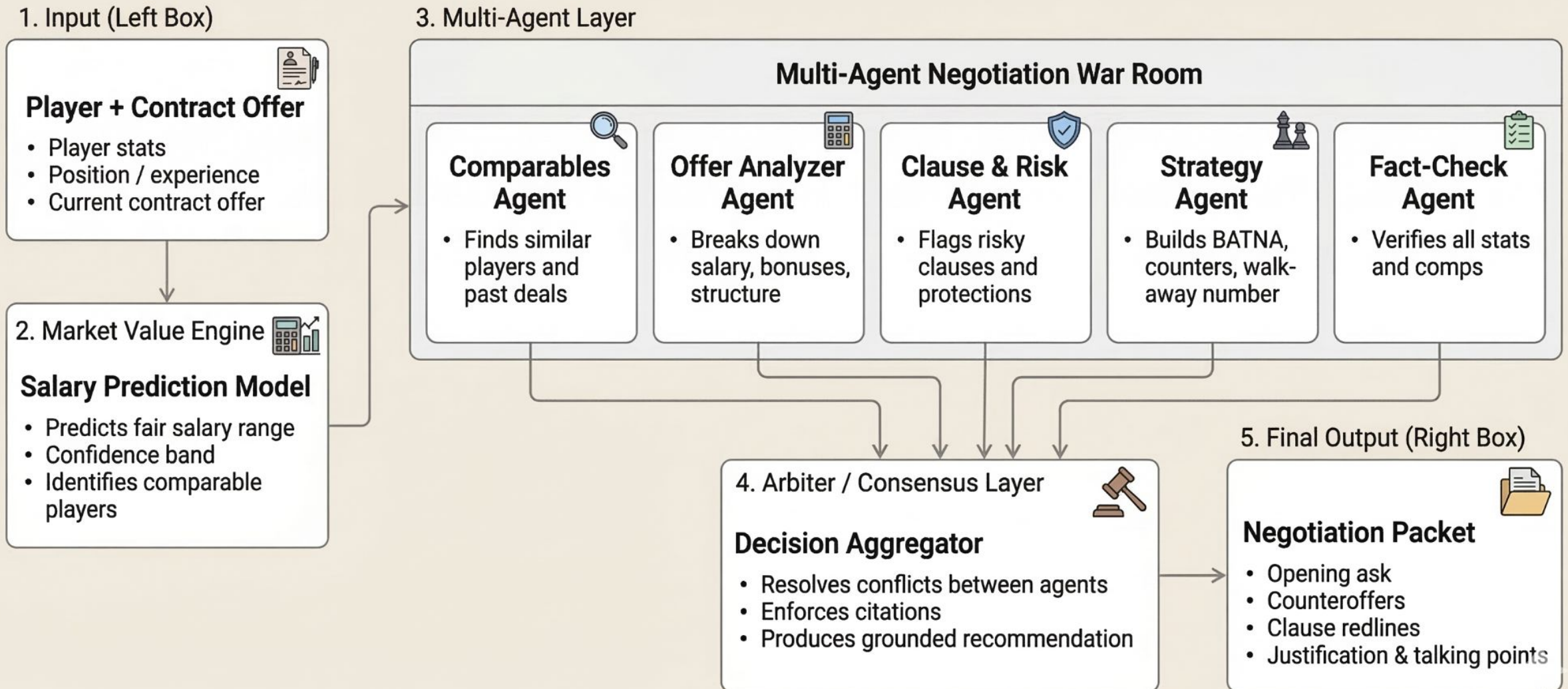
Constraint-Aware Deal Optimization

Agents evaluate salary caps, contract rules, market comps, and negotiation policies to iteratively propose legally compliant, market-aligned agreements.

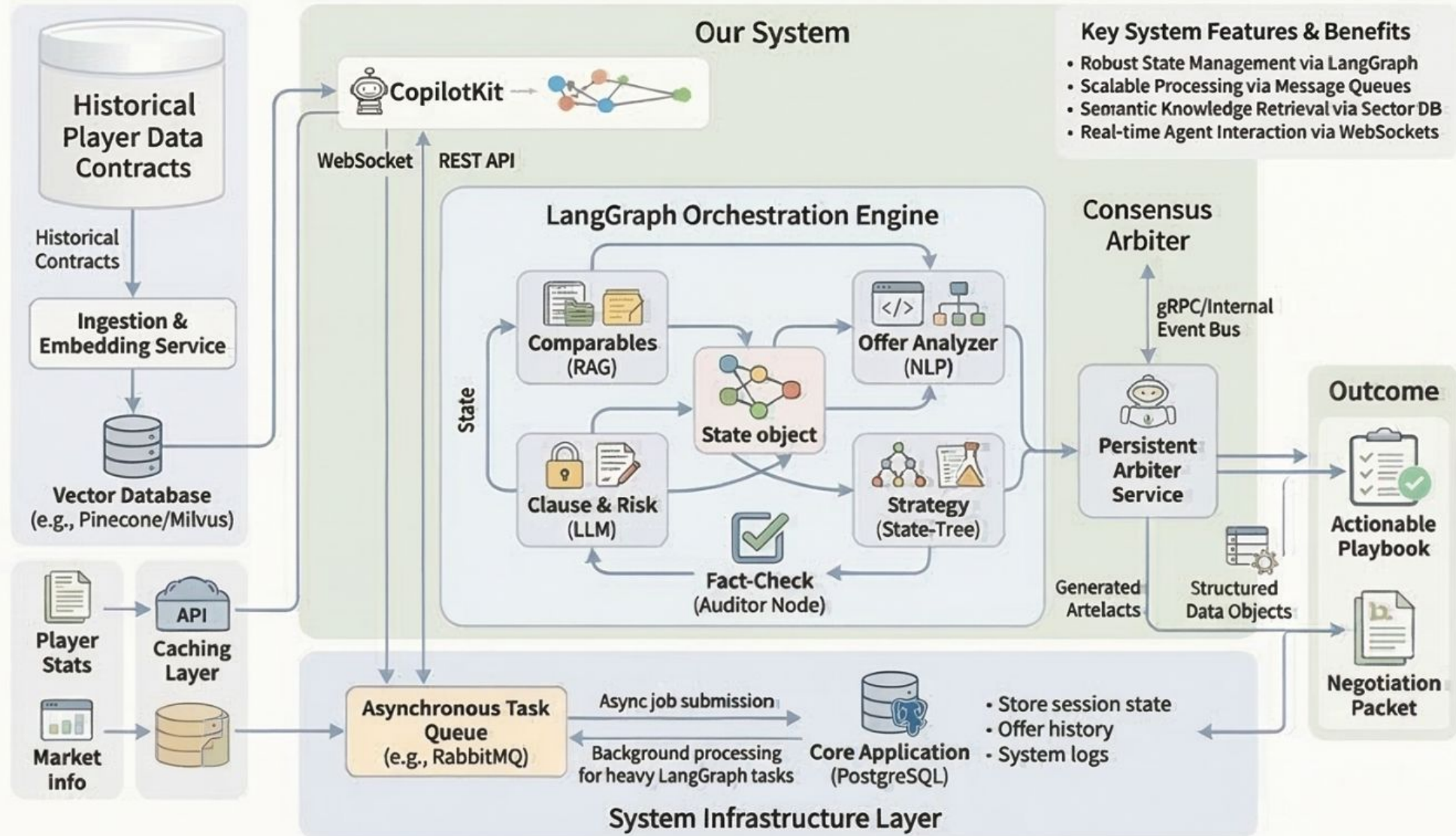
Autonomous Negotiation Workflow

CopilotKit interface enables users to define goals and constraints while the agent system conducts full negotiation cycles and produces final term sheets.

Modular Multi-agent Negotiation Architecture



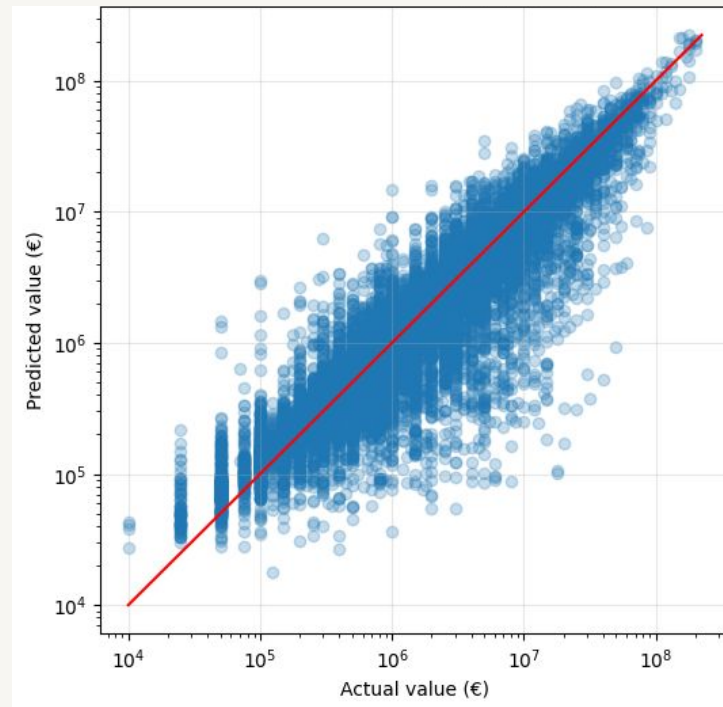
Modular Multi-agent Negotiation Technical Architecture



Agent 1: Salary Predictor

Some of the included features:

- Age at each valuation
- Club at each valuation
- Position and Sub-position
- Goals in the last 356 days
- Different valuations in the past
- Valuations changes(delta)



Current regression results:

R Square Score: 0.8332

Features importance (in order of importance):

- Amount of value change between valuations (delta)
- Age
- Minutes played in the last 180 days at each valuation
- Sub-position if goalkeeper or not
- Position if goalkeeper or not

Proposed next steps:

- Information about Leagues and if top tier or not
- Information about clubs such as their budget
- Tune regressor and do some features engineering

Constraint Checker Implementation

What it checks



Layer 0: Format

(valid JSON, required fields, numeric money units)



Layer 1: Contract sanity

(years bounds, non-negative, option year valid, incentives cap)



Layer 2: Market realism

(must fall within comps band $\pm$ tolerance)



Layer 3: Protocol rules

(turn order, max rounds, no repeat/cycle offers)

What it returns



**PASS/FAIL +
violation codes**

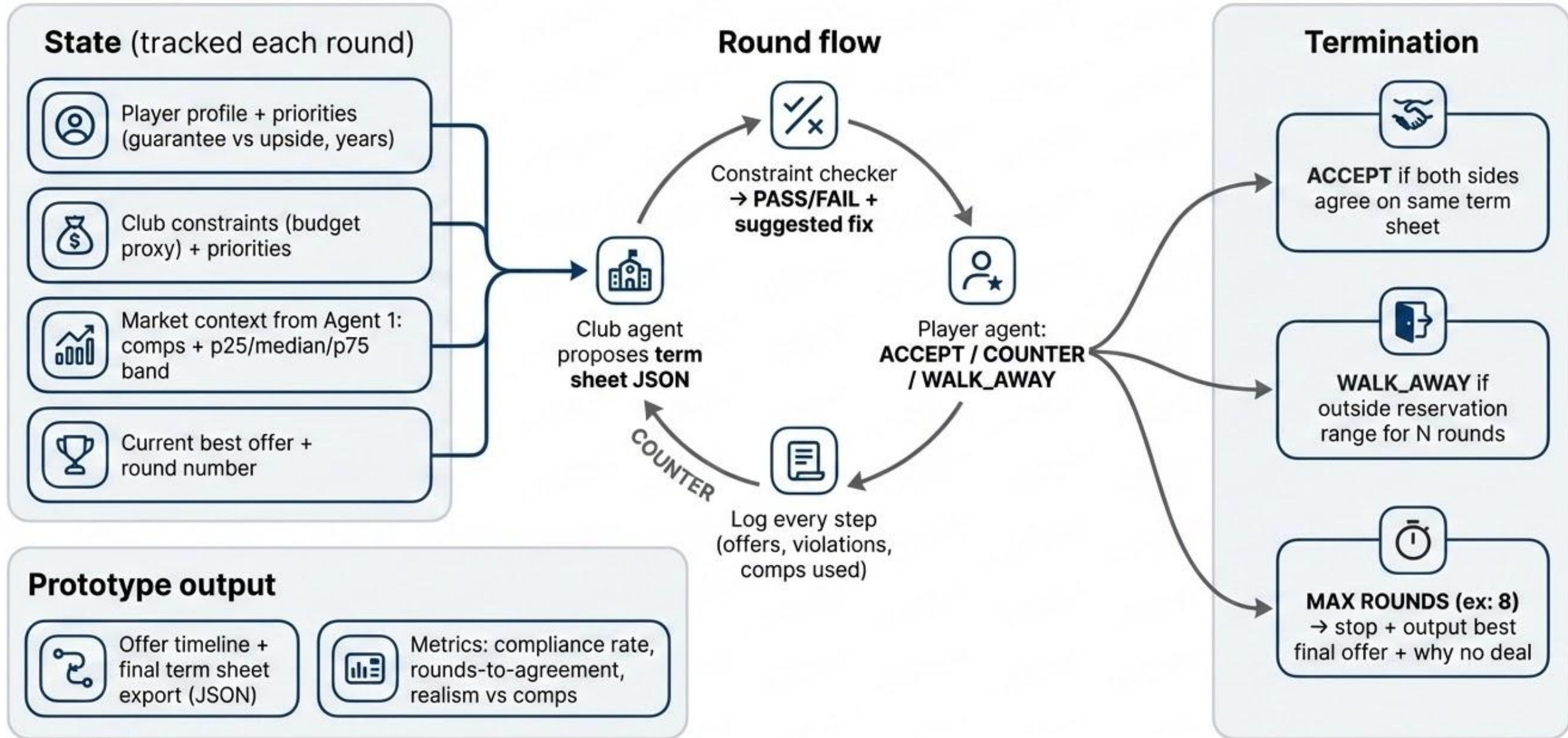


1-line explanation
(UI-friendly)



Suggested fix
(minimal edit to become valid)

Negotiation Loop + Termination Rules



In summary, our system combines machine learning, multi-agent negotiation, and constraint-based validation to create an autonomous negotiation assistant capable of producing realistic, data-driven contract proposals.



Thank you

Agent 2: Constraint Aware Deal Optimizer

Format + Schema

- Offer must be valid JSON with required fields only
- Correct types + units (money as numbers, no “\$25M” strings)
- Cross-field consistency (total comp = base + bonus + incentives)

Contract Validity

- Basic bounds: years range, non-negative values, option-year within contract
- Incentives sanity (cap incentives as % of base)

Market Realism (Data-grounded)

- Offer must stay within comps band (p25/median/p75 $\pm$ tolerance)
- Flags “too high / too low vs market” with clear reason

Constraint Checker Implementation

What it checks

- **Layer 0: Format** (valid JSON, required fields, numeric money units)
- **Layer 1: Contract sanity** (years bounds, non-negative, option year valid, incentives cap)
- **Layer 2: Market realism** (must fall within comps band $\pm$ tolerance)
- **Layer 3: Protocol rules** (turn order, max rounds, no repeat/cycle offers)

What it returns

- PASS/FAIL + **violation codes**
- 1-line explanation (UI-friendly)
- **Suggested fix** (minimal edit to become valid)

Negotiation Protocol:

- Max rounds (ex: 8) → then walk-away + best final offer summary
- Turn-taking rules (Club ↔ Player alternation)
- Limit invalid retries per turn (ex: 2 attempts)

Anti-loop Guardrails

- Reject identical repeated offers (hash term sheets)
- Detect offer cycles and force a concession or terminate

Deterministic Output (for agents + UI)

- PASS/FAIL + violation codes (e.g., OUTSIDE_COMPS_BAND, BUDGET_EXCEEDED)
- Human-readable explanation in 1 line
- Suggested fix: patched term sheet with minimal edits
- Next allowed actions: OFFER / COUNTER / ACCEPT / WALK_AWAY

Negotiation Loop + Termination Rules

State (tracked each round)

- Player profile + priorities (guarantee vs upside, years)
- Club constraints (budget proxy) + priorities
- Market context from Agent 1: **comps + p25/median/p75 band**
- Current best offer + round number

Round flow

- Club agent proposes **term sheet JSON**
- Constraint checker → PASS/FAIL + suggested fix
- Player agent: ACCEPT / COUNTER / WALK_AWAY
- Log every step (offers, violations, comps used)

Termination

- **ACCEPT** if both sides agree on same term sheet
- **WALK_AWAY** if outside reservation range for N rounds
- **MAX ROUNDS (ex: 8)** → stop + output best final offer + why no deal

Prototype output

- Offer timeline + final term sheet export (JSON)
- Metrics: compliance rate, rounds-to-agreement, realism vs comps



Thank you

Negotiation Loop + Termination Rules

